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DOI: 10.1111/1745-9133.70006

CRIMINOLOGY & Public Policy

ORIGINAL ARTICLE

Can enhanced street lighting improve public safety at scale?

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Funding information
National Institute on Minority Health and Health Disparities of the National Institutes of Health, Grant/Award Number: R01MD019163

Abstract

Research Summary: Street crimes are thought to be influenced by changes in ambient lighting; yet, most studies have focused on small-scale interventions in limited areas. It remains unclear whether enhanced lighting can improve safety on a larger, jurisdiction-wide scale. This study examined the first 10 months of Philadelphia's citywide rollout of enhanced street lighting. Between August 2023 and May 2024, 34,374 streetlights were upgraded across 13,275 street segments, converting roughly one third of the city's street segments to new LED (light-emitting diode) fixtures that provide clearer and more even lighting. We assessed the impact of these upgrades on total crimes, violent crimes, property crimes, and nuisance crimes. Our findings reveal a 15% decline in outdoor nighttime street crimes and a 21% reduction in outdoor nighttime gun violence following the streetlight upgrades. The streetlight upgrades may have contributed to 5% of the citywide decline in Philadelphia gun violence during the study period. Qualitative data also suggest that resident perceptions

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of safety and neighborhood vitality improved after new streetlights were installed.

Policy Implications: Although prior research has mostly focused on small-scale implementations, our study demonstrates that improvements in street lighting can significantly reduce crime rates across large urban areas. These results support the adoption of energy-efficient LED street lighting as a crime reduction strategy for cities. Further research is needed to explore the impact of enhanced streetlight interventions on other types of crime and to determine whether the crime reduction benefits are sustained when these upgrades are implemented across the entire City of Philadelphia for an extended period.

KEYWORDS

crime displacement, gun violence, place-based interventions, street lights

Serious crime is spatially concentrated within U.S. cities in the most socially disadvantaged communities (Weisburd et al., 2012). Even in the poorest neighborhoods though, most blocks experience few incidents of serious crime and gun violence each year, suggesting that, beyond poverty and social isolation, specific microgeographical features of the built and social environment play a key role in the concentration of crime (Braga et al., 2010; Sherman et al., 1989; Weisburd et al., 2004). Crime opportunity theories and a growing empirical literature in the social sciences suggest that small changes to place characteristics and situational dynamics can have an outsized impact on crime victimization (Braga & Clarke, 2014; Clarke, 1997; Wilcox & Cullen, 2018), including that which is perpetrated with guns. Prior work shows that a variety of place-based interventions to the built environment—including greening vacant lots, remediating blighted properties, and trash removal—can improve public safety and reduce gun violence and serious crime in disadvantaged communities (Branas et al., 2018; South et al., 2023).

Studies suggest that improving streetlights is an environmental change that can help increase public safety and reduce fear on city streets (Chalfin et al., 2021; Painter, 1996; Welsh et al., 2022). However, the available program evaluation literature on streetlights provides incomplete evidence of their impact on serious crime and gun violence because they tend to focus on a relatively small number of discrete locations (e.g., public housing complexes, parking lots, or neighborhoods). There is scant evidence on the public safety impact of improved street lighting throughout very large urban areas, so we do not know if enhanced lighting has the potential to have an impact on serious crime *at scale*. Indeed, a recently completed systematic review (Welsh et al., 2022) only identified one such study that evaluated the impact of an electrification program designed to improve street lighting coverage in rural Brazilian municipalities on homicide rates in those communities (Arvate et al., 2018). The lack of rigorous research on scaled-up interventions is a critical gap not only in research on street lighting but also in research on a wide array of crime reduction

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strategies (Braithwaite, 2021; Welsh et al., 2010). The gap is especially salient for policy makers who are searching for sustainable crime prevention strategies that will have at-scale impacts, not demonstration projects that provide limited proof-of-concept policy guidance.

Visibility on city streets can be further enhanced by replacing legacy streetlights with "smart" and energy-efficient LED lighting that reduces energy costs and carbon emissions. Brighter LED lights may also lead to crime control benefits relative to older high-pressure sodium (HPS) bulbs that emit lower lumen lighting with a yellowish hue (Wheeler, 2023). In this study, we evaluate the initial impact of Philadelphia's efforts to upgrade all city streetlights from HPS bulbs to LED lighting. Between August 2023 and May 2024, the city upgraded streetlight fixtures to smart[1] LED lighting at roughly one third of city street segments. As will be detailed below, we used difference-in-differences (DiD) estimators, paired with an event study model variant, to assess the effects of enhanced lighting on total crimes, violent crimes, property crimes, and nuisance crimes. Qualitative methods were used to capture the effects of lighting upgrades on visibility at treated streets and understand how residents of neighborhoods suffering from serious gun violence were impacted by streetlight improvements.

1 | PRIOR LITERATURE

Street lighting is an innovation that has supported the viability of urban life since antiquity. For instance, the Roman Empire used networks of oil lamps to provide ambient lighting at night in cities (Dossey, 2019). Candle-based streetlights were pioneered in the United States in the 18th century by Benjamin Franklin. By enhancing visibility, lighting has the potential to reduce crime by empowering potential victims to better protect themselves and by making potential offenders more aware that a public space has witnesses or that police are present (Farrington & Welsh, 2002). Aside from any immediate public safety benefits, improved nighttime lighting has been shown to reduce fear, making it generally popular among community residents (Painter, 1996). Therefore, an additional benefit to upgrading the lighting infrastructure is making areas more appealing to nighttime use, increasing neighborly socialization, and helping communities improve collective efficacy by producing more neighborly informal social controls (Kaplan & Chalfin, 2021; Painter & Farrington, 1997).

Streetlights are considered a key feature of the built environment that significantly impact public safety, including the prevention of gun violence victimization (Chalfin et al., 2021). As a result, cities often prioritize street lighting as a core strategy for improving public safety, exemplified by recent lighting upgrades in Los Angeles, Chicago, Dallas, and Philadelphia, among other cities. A comprehensive review of more than five decades of research by Welsh et al. (2022) found that street lighting interventions reduced crime by an average of 14% across 21 studies. Despite a wealth of quasi-experimental research on the crime prevention benefits of enhanced outdoor lighting, relatively few studies have employed rigorous designs that control for pre-existing differences in crime patterns between areas with upgraded streetlights and those without (Welsh et al., 2022). Additionally, much of the prior research has focused on a single neighborhood or a small number of neighborhoods, which complicates the accurate estimation of standard errors (Abadie et al., 2010).

Notably, the entire body of literature includes just one randomized controlled trial (RCT), conducted in NYC public housing. That study found that the installation of streetlights resulted in a reduction in felony assaults and robberies during nighttime hours (Chalfin et al., 2021), and subsequent work from this RCT found that enhanced street lighting led to reductions in both

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crime *and arrests* in the intervention areas that endured for at least 3 years (Mitre-Becerril et al., 2022). However, the lights studied in that intervention—portable temporary light towers—were markedly different from "normal" streetlights, thus creating uncertainty about the applicability of these findings to other settings. Likewise, although research shows that serious crime victimization is concentrated in public housing in NYC, the scale and size of high-rise public housing in NYC are unlike those of public spaces and residential areas in most other large U.S. cities. The study, therefore, does not tell policy makers what would happen to crime victimization if street lighting were enhanced in high-poverty neighborhoods consisting of low-rise buildings or single-family homes that are not as densely populated—a context where most of the urban poor live in the United States.

Critically, the RCT also does not reveal the extent to which a *citywide* improvement in street lighting will lead to a *citywide* reduction in crime victimization. Although this question continues to remain unknown in U.S. cities suffering from serious crime problems, empirical evidence from Brazil suggests that improved lighting can reduce homicide in rural areas of developing countries. Arvate et al. (2018) used instrumental variables and a regression discontinuity design to evaluate Brazil's Luz Para Todas ("Light For All") program and found that increased lighting in eligible municipalities was associated with a reduction of roughly 92 homicides on public streets and almost 18 homicides where death occurred in a hospital (per 100,000 inhabitants). Two other studies that rely on natural experiments offer powerful proof of concept that improvements to ambient lighting might be able to have an impact on crime at scale. The studies examine the effects of daylight-saving time (DST), when clocks are set forward by 1 hour, and find that the resulting increase in darkness is linked to a rise in street crimes, including those involving firearms (Doleac & Sanders, 2015; Dominguez & Asahi, 2017). Though the literature suggests that providing enhanced lighting to disadvantaged communities could yield significant public safety benefits, the intervention studied—the change in daylight from a celestial body—remains beyond the control of policy makers.

With respect to the scalability of lighting, several other recent lighting studies are noteworthy. A study of streetlight outages in Chicago addressed the small sample size limitation of the prior randomized experiment and much of the quasi-experimental literature by utilizing a natural experiment design, triggered by electrical outages that caused short-term disruptions to municipal street lighting (Chalfin et al., 2022). This research analyzed immediate changes in reported serious crimes and gun violence before, during, and after 300,000 streetlight outages. The study found that outages caused nighttime crime to spill over into nearby street segments, with the greatest impact on robberies and auto thefts. The results suggest that crime is sensitive to lighting but that blacked-out streets may prompt both victims and offenders to relocate to nearby streets with nighttime lighting, implying that reduced ambient lighting can lead to an increase in crime in better-lit locations.

A more recent study of the Oxfordshire, Reading, West Berkshire, and Wokingham municipalities in the United Kingdom examined what happened to crime after local authorities began to reduce the amount of street lighting at night to save energy costs and carbon emissions (Tompson et al., 2023). Consistent with the research of Chalfin et al. (2022) in Chicago, the study found that crime increased significantly on street segments adjacent to those shut off at midnight to save energy costs. The study also found that areas adjacent to streetlights that replaced yellow lights (HPS bulbs) with energy-efficient white lights saw nighttime crime decrease by 14% and increase by 13% on adjacent streets. Overall, however, they find no net change in crime, suggesting that switching lights off at midnight, dimming lights, or switching to more energy-efficient white lights simply reallocates crime nearby.

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The central theme across these studies is that discrete shifts in street lighting have led to spillover effects, raising concerns that piecemeal lighting upgrades may simply displace crime rather than produce a genuine, scalable improvement in public safety. Additionally, it is important to highlight previous research that examines a similar intervention—LED lighting upgrades—implemented at scale across an entire city. Wheeler's (2023) study, conducted in Dallas, Texas, focused specifically on firearm offenses. Using a triple-differences design that compares nighttime crimes to daytime crimes in areas with and without the LED lighting upgrades, the research found no significant effects on nighttime firearm offenses. However, it is crucial to note that despite the strong research design, the study was underpowered to detect an effect due to the statistical rarity of firearm offenses. In fact, 96% of the study's grid areas saw no lighting upgrades, and the average firearm offense rate was relatively low (0.12 per month).

Finally, several important limitations in the existing literature should be noted. First, much of the research is dated and predates the widespread use of technologies such as CCTV, smartphones, ShotSpotter, and other social and technological changes that have altered how public spaces are used and policed. Second, even the more recent studies, conducted between 2000 and 2020, took place during a period of broad crime decline. As a result, there is a lack of evidence about the effect of street lighting during periods when crime rates have risen significantly. Third, as highlighted by Chalfin et al. (2021), Mitre-Becerril et al. (2022), and Welsh et al. (2022) in their recent systematic review, there is some suggestive evidence that street lighting may have a modest impact on daytime crimes, potentially because it signals that an area is being cared for. However, this hypothesis remains debated and has yet to be tested on a large scale. Relatedly, due to the small sample sizes in most prior studies and the statistical rarity of crimes at the street segment level, the literature offers little insight into the specific conditions or locations within a city where improved lighting would be most effective (see Welsh et al., 2022).

1.1 | Theoretical framework

Crime opportunity theories, such as routine activity (Cohen & Felson, 1979), rational choice (Cornish & Clarke, 1986), and environmental criminology (Brantingham & Brantingham, 1991), are often used to understand the place characteristics, situations, and dynamics that cause crime events to concentrate at specific places. Routine activity theory suggests that crimes occur when likely offenders and suitable targets converge in space and time in the absence of a capable guardian (Cohen & Felson, 1979). Rational choice theory posits that offenders make rudimentary decisions to commit crimes based on their perceptions of the risks, effort, and rewards associated with the completion of a particular criminal act (Cornish & Clarke, 1986). Environmental criminology predicts that there will be more criminal offending in activity spaces that offenders frequent in edges of areas where there is a physical change in the land use and there is no clear ownership of spaces (Brantingham & Brantingham, 1981). These theoretical perspectives are applied to develop situational crime prevention interventions that seek to manipulate crime opportunity structures at specific places to make crime more difficult and riskier, and less rewarding and excusable, by prospective offenders (Clarke, 1997). These insights also support applied crime prevention through environmental design (CPTED) interventions (Jeffery, 1977), "defensible space" architectural practices (Newman, 1972), and problem-oriented policing programs (Braga, 2008; Goldstein, 1990). Improved street lighting is generally recognized as increasing the perceived risks of criminal acts for prospective offenders by enhancing natural surveillance of public spaces (e.g., see Clarke, 1997, p. 18). Better lighting may also stimulate stronger informal social

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control through enhanced place management and improved guardianship (Welsh & Farrington, 2009).

A large empirical literature in the social sciences suggests that small changes to the situational environment can have important effects on human behavior. With respect to the broader social sciences, this has been shown in the empirical impact of "nudges," where incremental environmental changes have a large impact on consumer behavior (Benartzi & Thaler, 2013; Madrian & Shea, 2001; Thaler, 2018; Thaler & Sunstein, 2009). With respect to the actions of potential offenders, numerous studies on situational crime prevention, CPTED, and other crime opportunity reduction measures suggest that changes to the built environment can influence criminal decision making, including the incidents of shootings, gun assaults, robberies, and other firearm offenses (Bowers & Guerette, 2014; Clarke, 1997; Cozens et al., 2005; Taylor & Gottfredson, 1986). These findings are, in part, explained by a body of empirical evidence in criminology as well as in economics that suggests that individuals most inclined to commit criminal offenses are present oriented and that they tend to place relatively little weight on future probabilities (Jacobs, 2010; Lee & McCrary, 2017; Nagin et al., 2015).

In a world in which potential offenders are actively contemplating criminal acts, small environmental design changes experienced in the present may well have an outsized impact on behavior relative to the uncertain prospect of prison sentences that, if experienced, will occur sometime in the future (Nagin, 2013; Nagin & Paternoster, 1994). For instance, qualitative research shows that individuals engaged in street-level robberies or outdoor drug dealing actively seek out dimly lit streets to conduct illegal activities (St. Jean, 2007; Wright & Decker, 1997). Gun violence and serious crime might therefore be deterred by tactical changes to the built environment, including a significant enhancement of streetlights.

Changes to the built environment can also have a strong effect on social cohesion. Multiple theories posit that visible environmental factors, including the lack of lighting on streets, lead to individuals being afraid to spend time on the streets (Painter, 1996). This negates the ability of residents to engage in surveillance of each other and maintain informal social controls that help mitigate antisocial behavior on the streets that can lead to gun violence (Sampson et al., 1997). In line with these theories, prior quasi-experimental and experimental studies show that place-based interventions decrease fear, increase neighborly socialization, and reduce gun violence outcomes. There are now a number of high-quality controlled trials and RCTs that show that availability of trees and green space, restoring vacant lots, and reducing trash, graffiti, and unkempt lots are associated with reductions in serious crimes and gun violence (Braga & Bond, 2008; Branas et al., 2011, 2016; Garvin et al., 2012; Kondo et al., 2018, 2015; South et al., 2023). These interventions share a focus on changing the situational environment to make areas less conducive to antisocial behavior, rather than by relying on police arrests or social service strategies. Research has similarly shown that improvements in lighting and other built environment features of convenience stores reduce robberies, in both an RCT and controlled trials (Casteel & Peek-Asa, 2000).

Direct changes to street lighting offer the potential benefit of disrupting situational aspects of places that lead to increased gun violence (e.g., street robberies, open-air drug dealing) without relying on the need for police to make arrests and widen the footprint of the criminal justice system. In the United States, public policy may therefore undervalue place-based interventions like street lighting as a method for reducing gun violence and serious crime, instead focusing on costly criminal justice inputs like police and prisons or resource-intensive social programs that often hinge on a charismatic leader or a highly specific programmatic model that are difficult to scale (Branas & MacDonald, 2014; Ludwig et al., 2011). In addition to prior work showing that place-based interventions can be effective approaches to reducing serious crime, there is

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also the empirical reality that serious crime is hyperconcentrated, suggesting that changes to the environmental aspects of places could be a particularly salient non-criminal justice approach to preventing crime.

2 | STUDY DATA AND METHODS

In the present study, we examine the first 10 months of the citywide rollout of enhanced street lighting in Philadelphia, Pennsylvania. We analyzed the effect of upgrading 34,374 streetlights across 13,275 blocks in Philadelphia between August 2023 and May 2024 on total crimes, violent crimes, property crimes, and nuisance crimes. The lighting fixture upgrades converted roughly one third of city street segments from HPS bulbs to LEDs. The blocks on which the initial lighting upgrades were completed are disproportionately among the higher crime blocks in the city. We utilized crime data from July 2022 to May 2024 such that we have a minimum of 12 months of data on crime before any streetlight upgrades were made. A DiD design, paired with a variant on an event study, was used to examine the effect of enhanced smart streetlights on serious crimes across a range of neighborhood contexts.

Our study also includes a qualitative component to examine residents' experiences and to understand how communities most impacted by gun violence and serious crime may be affected by streetlight improvements. As will be described below, we conducted 20 hours of ethnographic observations of post-upgrade street visibility, completed 19 semistructured interviews, and convened four resident focus groups in selected areas. Qualitative data collection was carried out in partnership with the Urban Affairs Coalition (UAC).

2.1 | Study context

The Philadelphia Energy Authority (PEA) is leading one of the most ambitious public infrastructure projects in the United States, a project that will, upon its completion, replace all 134,552 streetlights in the city of Philadelphia. In August 2023, the project began upgrading the city's entire lighting network to energy-efficient, "smart" LED lights. This initiative involves replacing the city's existing legacy yellow bulbs with LEDs, with the goal of completing the installation by the end of 2025. Although the new LED lights are not necessarily brighter in terms of lumens, they offer better quality lighting and more even coverage of an area, greatly reducing the number of "dark spots" in the community. The white light emitted by the LEDs will also appear brighter compared to the legacy HPS bulbs that have been in place for more than a generation (Kim et al., 2015). Although the typical LED lights use significantly less energy, they improve overall street illumination. Figure 1 provides a visualization of streets in Southwest Philadelphia and shows a contrast in lighting. The street in the image on the left features the legacy HPS streetlights stretching into the distance, whereas the street in the right image features the upgraded LED streetlights extending as far as the eyes can see.

Given concerns about the effects of LED lighting that is too bright on human health, the PEA consulted with neighborhood residents and many of the lights have a color temperature of less than 3000K, which is consistent with the American Medical Association's policy statement on LED lighting.[2] The smart lighting initiative can also potentially cut down on the length of streetlight outages by automatically notifying city planners when a light is out, rather than relying on

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FIGURE 1 Contrast between streets with HPS bulbs (Left) LED lighting (Right). [Color figure can be viewed at wileyonlinelibrary.com]

citizen 311 calls, which could be substantially delayed if residents assume that the local energy authority is keeping track of lights that are not working.

2.2 | Quantitative data

We capture changes in public safety using publicly available microdata on crimes and shootings reported daily to the Philadelphia Police Department (PPD), augmented with an additional PPD data element that allows us to observe whether a crime occurred indoors or outdoors. Using open data sources (opendataphilly.org/dataset/shooting-victims; opendataphilly.org/dataset/crime-incidents), each incident includes the date and time of the crime, as well as the geographic coordinates (x–y) coded to the nearest block. The shooting incidents that the police collect have been corroborated with records found in hospital data for Philadelphia (Branas et al., 2008). Lighting installation data provided by the PEA include the exact date and geographic coordinates (x–y) of light installations starting in August 2023 and ending in May 2024. All geographic coordinates were converted into Mercator projection prior to conducting any of the spatial analyses.

To determine whether crimes occurred during the day or night, we follow previous research that used data on civil twilight hours to determine when natural sunlight was not available. Civil twilight typically begins about 30 minutes after sunset and ends around 30 minutes before sunrise. During this time, there is enough light to clearly see objects, whereas after these hours, street lighting is usually needed to clearly see outdoor activities. We obtained data on civil twilight hours from the United States Naval Observatory (USNO), which provides exact times for when civil twilight starts and ends in Philadelphia each day. Since victimization data often do not include the exact time of the event, we classified crimes as happening at night if the incident was reported during the civil twilight period.

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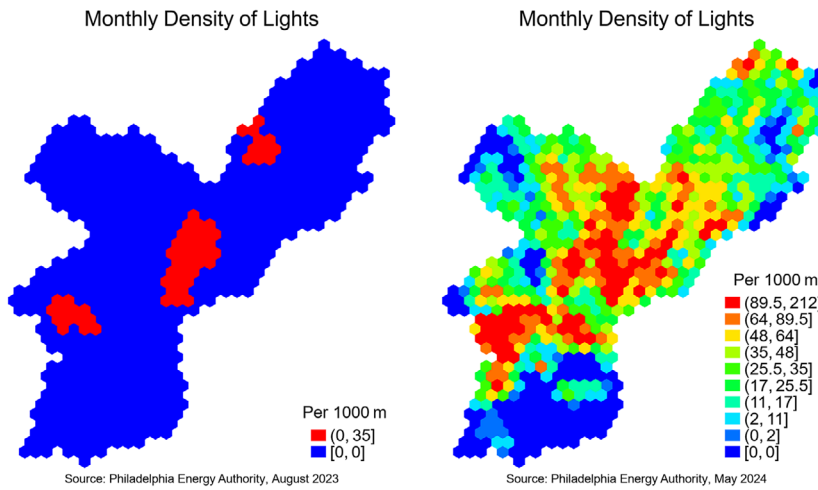

FIGURE 2 LED density during first (left) and last (right) months of observation. [Color figure can be viewed at wileyonlinelibrary.com]

Data were analyzed at two levels: the street segment level and a larger, aggregated area—hexagons that cover 1000 meters in width. The latter approach allows us to study the impact of lighting while accounting for potential spatial spillovers. Following a common method in place-based crime research (Ratcliffe, 2010), we divided Philadelphia into 720 hexagons, each covering roughly 0.51 square kilometers. For each of these hexagons, we calculated the number of crimes and the cumulative number of streetlight fixtures that had been updated as of each month. Borrowing an approach used by Branas et al. (2018), we accounted for the distance between a given crime and an upgraded light fixture by calculating a monthly kernel density estimate of the rate of crime and upgraded light fixtures each month per square meter in each hexagon between June 2022 and May 2024.[3] Kernel density creates a smoothed estimate of the distance from the center point of each hexagon to every crime or upgraded LED fixture, where crimes and upgraded lights are given greater weight when they are closer in distance to the center point of each hexagon. We used a bandwidth of 1000 meters to encompass the entire hexagon along with edge correction.[4] Figure 2 shows the kernel density estimates for lights during the period of the study. For dependent variables, we used the total reported crimes and shootings, and generated subcategories of gun violence, violent, property, and nuisance crimes as well as a placebo category composed of financial crimes. Gun crimes represent the combined total of reported shootings, gun robberies, gun assaults, and weapons violations. Violent crimes include the total of reported rapes, assaults (aggravated, simple, and other), murder (felony), robbery (both firearm and nonfirearm), and shootings that did not result in a murder. Property crimes include the total of burglaries (both commercial and residential), thefts (such as motor vehicle thefts and thefts from vehicles), and vandalism or criminal mischief. Nuisance crimes include the total of reported disorderly conduct, public drunkenness, liquor law violations, and narcotics or drug law violations. Financial crimes include the total of fraud, forgery, and embezzlement.

We recorded the total number of each crime type in each hexagon-month, including daytime and nighttime crimes, using civil twilight hours to distinguish between the two. Given that time stamps on crime may reflect delayed victim reporting rather than when the crime occurred, we conducted a robustness check and removed the 2-hour buffer after civil twilight began each day

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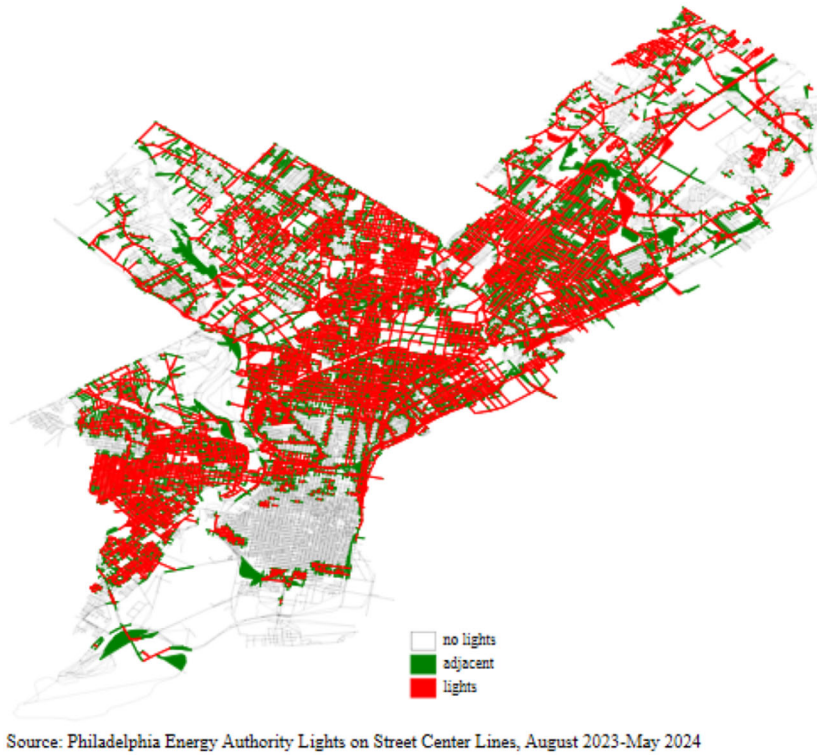

FIGURE 3 Upgraded streetlights and adjacent street segments. [Color figure can be viewed at wileyonlinelibrary.com]

and re-estimated our models. To focus on the direct impact of lighting on outdoor crimes, we also re-estimated the models using an indicator for whether the crime occurred indoors or in an outdoor public space. These location designations (e.g., outside street/highway) were based on descriptions provided by the PPD and are available through a link to a publicly available unique crime event identifier. Although lighting is expected to be most effective at night and for outdoor crimes, it is important to recognize potential measurement errors, including delays in civilian reports and inaccuracies in the reported location of crimes. We address this concern in our empirical analysis.

In separate analysis to assess potential crime displacement and diffusion effects caused by the installation of upgraded streetlights, we geocoded each upgraded light and each crime to its nearest street segment in Philadelphia. We then calculated the number of crimes for the same categories and new upgraded streetlights on each segment per month between July 2022 and May 2024. Figure 3 shows the layout of street segments with lights installed and those adjacent to them in Philadelphia.

2.3 | Models

To estimate the effect of enhanced streetlights on crime and gun violence, we use a DiD research design, paired with an analysis that tests the estimator's common trends assumption. The analysis leverages the staggered adoption of the improved streetlights. Our two-way fixed effects

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(TWFE) regressions—which we augment with the more novel approach to DiD estimation of de Chaisemartin and D'Haultfœuille (2020)—take the following form:

$$Y_{it} = \alpha + \beta \text{Lights}_{i(t)} + \delta_i + \pi_t + \varepsilon_{it}. \tag{1}$$

We estimate (1) with a least squares regression model specification where the dependent variable, Y_{it}, is the number of crimes that occurred within hexagon grid i during year-month t. $\text{Lights}_{i(t)}$ is a measure of the density of enhanced lighting in a given hexagon grid i per month (t). Fixed effects for each hexagon and year-month are included, and standard errors are clustered at the hexagon level to account for dependence in the errors within hexagons throughout time. The coefficient, β, thus estimates the average effect of upgrading an additional new light fixture in a hexagon on the number of monthly crimes per kilometer. The key assumption for interpreting our estimate of β as a causal effect is the parallel trends assumption. This means that, in the absence of enhanced lighting, crime trends in areas that receive lighting and those that are yet to receive upgraded lights would have been identical if the lighting upgrades were not installed. Following best practices, we examine the parallel trends in the months leading up to the implementation of enhanced lighting (MacKinlay, 1997; Roth, 2022).

Recent literature has noted two pressing concerns with the use of TWFE models to estimate DiD treatment effects (de Chaisemartin & D'Haultfœuille, 2020). First, TWFE models are partially identified based on a comparison between later treated locations and earlier treated locations that have already received the treatment. When treatment effects are heterogeneous—which is typically the case—the result is that TWFE models will generate a biased estimate of the treatment effect. Second, TWFE estimates are sensitive to variation in the timing of treatment across units, giving greater weight to units that are treated in the middle of the observation window. To address this issue, we rely on a heterogeneity-robust DiD estimator that controls for the fact that streetlight upgrades in each hexagon enter the study at different periods and is a continuous treatment (de Chaisemartin & D'Haultfœuille, 2020). We estimate the following model:

$$Y_{it} = \alpha + \delta_i + \pi_t + \mu G_i + \beta \text{Lights}_{i(t)} + \epsilon_{it}, \tag{2}$$

$$\hat{\beta}_{\text{DiD}} = \frac{\sum_{i,t} \left(Y_{it} - \overline{Y_t} \right) \times \left(\text{Lights}_{i(t)} - \overline{\text{Lights}_t} \right) \times G_i}{\sum_{i,t} \left(\text{Lights}_{i(t)} - \overline{\text{Lights}_t} \right)^2}.$$

In the DiD setup, G_i indicates period-specific groups (year-month) of upgraded lights and accounts for the differential exposure to light upgrades by time. We rely on the Stata program developed by de Chaisemartin et al. (2019), which compares the period-specific groups to placebo groups where the timing of light upgrades is randomized. The model then estimates the average total adjusted effect across groups ($\hat{\beta}_{\text{DiD}}$).

With place-based interventions, there is always a concern that the intervention will displace crime by simply pushing it around the corner. The kernel density estimates guard against this problem by providing greater weight to crimes and light locations that are near each other. However, in keeping with prior literature, we also examined the impact of the streetlight intervention on crimes in adjacent street segments to see if enhanced lighting on a street segment leads to crime rising in adjacent street segments. We define adjacent segments based on whether a street

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segment is 0–10 feet from a street that receives enhanced lighting. To ensure that each street segment receiving an enhanced streetlight is compared to a unique adjacent segment, we choose a randomly selected street segment that is within 10 feet, with no adjacent segments belonging simultaneously to more than one treated area. For this analysis, we rely on counts of crime and light density per street segment in each month and estimate a DiD Poisson regression according to the following form:

$$Y_{it} \sim \text{Poisson}\left(\lambda_{it}\right),$$

$$Y_{it} = \exp\left(\alpha + \beta \text{Lights}^{\text{adj}}_{i(t)} + \theta_i + \pi_t\right). \tag{3}$$

In Equation (2), Y_{it} is the number of crime outcomes that occurred on a street segment i during a month (year-month) t in segments adjacent to the upgraded lighting areas relative to street segments that are farther away. In this model, Lights is a measure of the running count of enhanced lighting installed in the nearby street segment in each month (t). Fixed effects for each street segment (θ) and year-month (π) are included to control for segment- and time-specific effects. The coefficient β thus estimates the average effect of additional new lighting on crime per month in street segments adjacent to the lighting upgrades relative to street segments further away. Thus, this model provides insight into how the lighting intervention on street segments influences crime on the adjacent streets.

2.4 | Qualitative methods

To examine whether community members and key stakeholders felt that the enhanced streetlights modified the social dynamics of their neighborhoods, we conducted ethnographic observations, semistructured interviews, and focus groups with residents in four sections (Districts) of the city with high rates of gun violence where upgraded streetlights were installed (Northwest, Northeast, West, Southwest). Qualitative data collection occurred between March 2024 and September 2024. During this period, UAC community partners assisted the research team in identifying key times and areas for ethnographic observations, as well as recruiting participants and organizing focus groups.

Ethnographic observations were completed during the pre-intervention (i.e., when older HPS lighting was present) and intervention periods (i.e., after installation of new LED lights) in each of the four city sections (eight total observation sessions). Observers were provided with route maps through clusters of blocks that were scheduled to get new LED lights before and after the streetlights were installed. Each ethnographic observation lasted for approximately 1 hour and was conducted during evening hours (starting about 20 minutes before sunset and ending 20 minutes after) to assess street visibility and capture changes in neighborhood activity as streetlights were turned on. During the intervention period, the research team held focus groups with six to eight residents in each of the four city sections (a total of 28 residents across the four meetings) and conducted semistructured interviews with 19 residents (four to five per area). Focus group discussions and interviews generally followed structured topics covering neighborhood conditions, community cohesion, general perceptions of fear and public safety, experiences using public places, gun violence, and support for enhanced lighting in their neighborhoods.

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Data analysis followed a focused coding approach aimed at directly answering the research question: *How do streetlight upgrades impact community safety perceptions and behavior in high gun violence neighborhoods?* We used Saldaña's (2015) two-cycle coding process to analyze interview transcripts, focus group transcripts, and ethnographic field notes within NVivo 14 software. The initial coding phase involved a thorough examination of all data sources to identify emerging themes, whereas the second cycle concentrated on systematically organizing codes around core categories relevant to the research question. Analysis was conducted concurrently with data collection, enabling the research team to refine focus areas based on emerging concepts and insights from direct field experiences and participant interactions. To ensure reliability, the team revisited interview and focus group transcripts once primary themes were established, paying particular attention to prevalent patterns and deviant cases. Participant quotes were lightly edited for clarity while maintaining their original meaning, providing concrete examples of identified themes and subthemes. Pseudonyms are used below when study participants' quotes are reported. This analytical process facilitated a nuanced understanding of the ways streetlight upgrades affect community safety perceptions and behaviors in these high-risk neighborhoods.

3 | RESULTS

3.1 | Hexagon crime rates

Table 1 presents the average rate of overall crime and by category for the period before any lights were installed in Philadelphia. Means and standard deviations are presented for the entire study sample as well as for areas that received upgrades during the study period and those that did not receive upgrades. There were, on average, 13.6 total crimes per kilometer in Philadelphia per month between July 2022 and May 2024. Areas with lights installed between August 2023 and May 2024 ($n = 627$) have, on average, higher rates of crime than areas where lights have yet to be installed ($n = 93$) during the 23-month period. However, for the purpose of this study, we are interested in the relative change in crime in areas before and after light upgrades were installed. The key to our method is that these patterns of crime change throughout the period from the beginning of light installations in August 2023 to the end of our observation window in May 2024, at which point roughly one third of street blocks in Philadelphia have upgraded LED lights.

Table 2 presents the results from our primary regressions on the effects of changes in the dosage of LED streetlights per hexagon, both overall and by outdoor and indoor locations. The table reports the point estimates and standard errors for the effect of upgrading each additional light. For ease of interpretation, we transform the coefficient by multiplying it by the average number of lights upgraded to compute a percentage change in crime resulting from the intervention. We report the percentage change in crime resulting from the intervention, along with a 95% confidence interval for this estimate.

Lighting upgrades are associated with a significant reduction in overall crime as well as gun crimes, violent crimes, and property crimes. The results indicate that the average increase in LED lighting among the 720 hexagons in the city led to a 5% decrease in overall crime, a 4% decrease in violent crime, a 7% decrease in property crimes, and a 17% decrease in gun crimes. There is no impact of upgraded streetlights on nuisance crimes, which tend to be police-discovered rather than to be reported by residents. Consistent with expectations, there is no impact of streetlight upgrades on financial crimes—crimes that are very likely to be committed indoors. For gun

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TABLE 1 Comparison of areas by upgraded lights.

	Mean	*SD*	Min	Max	*N*
No. upgrades					
Total crimes	3.07	7.64	0.0	48.6	2139
Gun crimes	0.09	0.30	0.0	2.9	2139
Violent crimes	0.65	1.65	0.0	12.2	2139
Property crimes	1.94	5.08	0.0	33.7	2139
Nuisance crimes	0.02	0.11	0.0	1.7	2139
Financial crimes	0.23	0.65	0.0	5.9	2139
Light upgrades					
Total crimes	15.17	15.55	0.0	174.8	14,421
Gun crimes	0.85	1.30	0.0	15.7	14,421
Violent crimes	3.80	4.42	0.0	36.4	14,421
Property crimes	8.35	8.77	0.0	120.7	14,421
Nuisance crimes	0.31	1.41	0.0	54.7	14,421
Financial crimes	0.73	0.82	0.0	19.7	14,421
Entire sample					
Total crimes	13.61	15.32	0.0	174.8	16,560
Gun crimes	0.76	1.25	0.0	15.7	16,560
Violent crimes	3.40	4.30	0.0	36.4	16,560
Property crimes	7.52	8.66	0.0	120.7	16,560
Nuisance crimes	0.28	1.32	0.0	54.7	16,560
Financial crimes	0.66	0.82	0.0	19.7	16,560

Note: Lights installed between August 2023 and May 2024. N = hexagon × 23 months.

crimes, the overall counts are relatively rare, so there is a larger confidence interval (−21.3% to −13.1%), but the effects are larger in magnitude than for any other crime subcategory.

Table 2 also highlights that the effects of streetlight upgrades on crime are primarily driven by crimes occurring outdoors. Specifically, lighting upgrades resulted in a statistically significant 15% reduction in overall outdoor crimes, a 19% decrease in gun crimes, an 11% decline in violent crimes, and a 16% decline in property crimes. In contrast, no statistically significant effects were found for indoor crimes, and the 95% confidence intervals rule out even modest changes. The only exception is gun offenses, where a statistically significant reduction in indoor gun crimes was observed. Overall, these findings align with the expectation that enhanced street lighting leads to reductions in criminal activity, particularly in outdoor areas.

Next, we examine daytime and nighttime crimes separately. Although enhanced lighting is expected to have the greatest impact on the amount of available lighting at night, prior literature suggests that daytime effects are possible as well (Mitre-Becerril et al., 2022; Welsh et al., 2022). We therefore report estimates for daytime and nighttime crimes, as well as outdoor crimes in both periods. Table 3 presents the results of this analysis.

During nighttime hours, gun crimes declined by 20% due to the lighting upgrades, whereas violent and property crimes decreased by 12% and 10%, respectively. The reduction in nighttime outdoor crimes was particularly pronounced, with an overall decline of 15%, including a 21% drop in gun crimes and a 20% and 16% decrease in violent and property crimes, respectively. Consistent

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TABLE 2 Effects of changes in lights on crime.

	Total	Gun	Violent	Property	Nuisance	Financial
All areas	−0.0287***	−0.00519***	−0.00610***	−0.0208***	0.000307	0.000307
	(0.00419)	(0.000632)	(0.00142)	(0.00321)	(0.00153)	(0.000420)
Ave. Post	15.34	0.825	3.929	8.146	0.385	0.722
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−6.605	−21.29	−6.214	−9.132	−19.05	−1.964
% Change	−5.138	−17.22	−4.267	−7.018	2.193	1.172
Upper 95%	−3.669	−13.14	−2.317	−4.900	23.37	4.304
Indoor	0.00510	−0.00127***	−0.000137	0.00408	0.000350	0.000368
	(0.00297)	(0.000227)	(0.000861)	(0.00269)	(0.00021)	(0.000371)
Ave. Post	7.251	0.184	2.209	3.834	0.0553	0.629
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−0.270	−25.29	−2.271	−0.859	−3.269	−1.573
% Change	1.935	−18.76	−0.170	2.927	17.31	1.607
Upper 95%	4.137	−12.19	1.931	6.702	37.59	4.782
Outdoor	−0.0386***	−0.00417***	−0.00674***	−0.0253***	−0.000372	0.0000176
	(0.00297)	(0.000572)	(0.000900)	(0.00195)	(0.00127)	(0.000117)
Ave. Post	7.161	0.604	1.636	4.224	0.306	0.0685
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−16.96	−23.92	−14.23	−18.88	−25.49	−8.469
% Change	−14.75	−18.89	−11.29	−16.42	−3.351	0.706
Upper 95%	−12.54	−13.84	−8.344	−13.95	18.89	9.872

Note: Standard errors in parentheses; Hexagon and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

with the theory that lights would impact crimes at night, Table 3 shows that the lighting upgrades resulted in larger crime reductions during nighttime hours compared to daytime hours, with overall crimes declining by 8% at night versus 4% during the day. Yet, it is noteworthy, consistent with prior literature, that outdoor daytime crimes also decreased modestly. We do, however, caution that the time stamps for property crimes in the public crime data may be particularly noisy due to reporting delays. The observed 17% reduction in outdoor daytime property crimes could reflect victims reporting overnight thefts in the morning, with the precise timing of the crime likely unknown.

3.2 | Street segment crime rates

Table 4 examines the relationship between the number of upgraded lights and crime on adjacent street segments relative to those further away. We examine the effect overall and at night given that counts of crime per street segment are relatively rare in each month. The results show that upgraded streetlights cause crime to decline slightly on adjacent street segments relative to those further away. The effects suggest that streets with upgraded lights cause crime to decline slightly

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TABLE 3 Effects of changes in lights on daytime versus nighttime crime.

	Total	Gun	Violent	Property	Nuisance	Financial
Day: Total						
	−0.0142***	−0.00216***	0.000111	−0.0132***	0.000164	0.000199
	(0.00342)	(0.000384)	(0.00108)	(0.00284)	(0.00107)	(0.000390)
Ave. Post	10.58	0.403	2.468	6.004	0.253	0.625
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−5.419	−19.80	−2.239	−8.571	−20.89	−2.486
% Change	−3.681	−14.72	0.124	−6.030	1.788	0.876
Upper 95%	−1.941	−9.620	2.487	−3.485	24.41	4.235
Day: Outdoors						
	−0.0259***	−0.00168***	−0.00205***	−0.0188***	−0.000411	0.0000329
	(0.00238)	(0.000352)	(0.000604)	(0.00167)	(0.000858)	(0.000113)
Ave. Post	4.883	0.279	0.992	3.116	0.195	0.0582
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−17.12	−23.16	−8.954	−19.39	−29.27	−8.868
% Change	−14.53	−16.46	−5.686	−16.54	−5.794	1.555
Upper 95%	−11.93	−9.713	−2.409	−13.68	17.86	11.96
Night: Total						
	−0.0145***	−0.00302***	−0.00621***	−0.00764***	0.000139	0.000111
	(0.00200)	(0.000367)	(0.000666)	(0.00119)	(0.000479)	(0.000116)
Ave. Post	4.758	0.422	1.461	2.142	0.132	0.0966
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−10.61	−24.17	−14.08	−12.76	−16.50	−3.349
% Change	−8.367	−19.55	−11.65	−9.781	2.887	3.145
Upper 95%	−6.116	−14.92	−9.204	−6.800	22.20	9.616
Night: Outdoors	−0.0127***	−0.00250***	−0.00469***	−0.00651***	0.0000353	−0.000013
	(0.00101)	(0.000318)	(0.000477)	(0.000570)	(0.000430)	(0.000037)
Ave. Post	2.277	0.325	0.644	1.108	0.111	0.0104
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−17.58	−26.17	−23.82	−18.83	−19.89	−22.93
% Change	−15.21	−20.98	−19.89	−16.09	0.876	−3.664
Upper 95%	−12.84	−15.77	−15.94	−13.34	21.62	15.69

Note: Standard errors in parentheses; Hexagon and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

in their adjacent street segments. The incident rate ratios show that crime drops by a total of 1.3% per month for an additional light upgrade to a street segment. Given the average number of lights is 2.5 per block segment (min = 0, max = 40), this suggests at most a 3.25% reduction in total crime caused by the average increase in lights on an adjacent street segment relative to street segments further away. The effects are similar for violent and property crimes. These estimates imply that some of the overall crime reductions caused by upgraded streetlights are the result of spillover

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TABLE 4 Effects of changes in lights on crime in adjacent versus nonadjacent street segments.

All	Total	Gun	Violent	Property	Nuisance	Financial
Lights	0.987**	0.996	0.985**	0.986*	0.985	1.002
	(0.00439)	(0.00672)	(0.00464)	(0.00653)	(0.00779)	(0.00852)
Ave. Dep	0.368	0.0710	0.144	0.239	0.106	0.0668
N	499,169	125,787	307,786	441,416	29,992	143,865
Day: Lights	0.986**	0.999	0.987**	0.984*	0.978*	1.004
	(0.00446)	(0.00863)	(0.00490)	(0.00654)	(0.00888)	(0.0090)
Ave. Dep	0.280	0.0602	0.112	0.196	0.103	0.0647
N	471,615	76,360	250,309	410,159	21,505	132,480
Night: Lights	0.992	0.993	0.982**	0.993	1.010	0.987
	(0.00561)	(0.00933)	(0.00664)	(0.00832)	(0.0202)	(0.0217)
Ave. Dep	0.153	0.0576	0.0854	0.104	0.0701	0.0481
N	339,181	75,302	188,945	239,683	13,616	21,689

Note: Robust standard errors in parentheses; block and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

effects on nearby areas. We also examined the estimates when we include the count of upgraded lights as a factor variable, and the results show that the effects are driven by more upgraded lights on a block (see Table A2). This again suggests that a meaningful change in upgraded lights on a street segment likely leads to a small diffusion of crime reduction nearby. The results are consistent with research in criminology that suggests that place-based interventions sometimes lead to a diffusion of benefits (Guerette & Bowers, 2017; Weisburd et al., 2006).

The results in Table 4 show that the decrease in crime on adjacent street segments does not differ by nighttime and daytime criminal activity, but the count of crime on street segments is sufficiently small so we cannot estimate differences between groups with any precision. These results are consistent with the hexagon analysis in suggesting that lights are not simply displacing crime to another location.

3.3 | Robustness

To ensure that our primary estimates, reported in Table 2, are not affected by a "forbidden comparison" or by arbitrary weights imposed by TWFE estimators, we present heterogeneity-robust DiD estimates (de Chaisemartin & D'Haultfœuille, 2020) that account for the variation in the number of months each area has had upgraded lights. The results, presented in Table 5, indicate significant reductions in total crime, gun crimes, violent crimes, and property crimes, with effects that are similar in magnitude to those in Table 2. The estimates, averaged across 3.28 months, suggest a reduction of 0.023 crimes per kilometer, compared to 0.028 in our primary estimates in Table 2. Additionally, there is a small, statistically significant increase in nuisance crimes, along with a decrease in financial crimes. Overall, these estimates, which account for varying upgrade timing, support the conclusion that our primary results are not driven by the specific periods when the upgrades occurred.

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TABLE 5 Heterogeneity estimates of lighting upgrades on crimes.

	Total	Gun	Violent	Property	Nuisance	Financial
Overall:						
Total effect	−0.0692***	−0.0074***	−0.00988***	−0.0568***	0.00153***	−0.00222***
	(0.00496)	(0.000503)	(0.00122)	(0.00369)	(0.000203)	(0.000559)
Outside:						
Total effect	−0.0629***	−0.0054***	−0.0109***	−0.0491***	0.00160***	−0.000546***
	(0.00391)	(0.000421)	(0.000843)	(0.00318)	(0.000161)	(0.000141)
Night: Outside						
Total effect	−0.00433***	−0.0022***	−0.00337***	−0.00592***	0.00229***	−0.0000235
	(0.00104)	(0.000294)	(0.000519)	(0.000609)	(0.000101)	(0.0000399)
Observations	6936	6936	6936	6936	6936	6936
N (switch × months)	4270	4270	4270	4270	4270	4270
Average months	3.28					

Note: Standard errors in parentheses; Hexagon and year-month fixed effects. Total effect is the total across 3.2 months.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

Next, we assess how sensitive the results are to the assumption of parallel trends, or that the trends in crime in areas with and without upgraded lights would have followed similar trajectories without the upgraded lights. We begin by estimating an event study, assessing whether hexagons receiving the first lighting upgrade were on a parallel path to other hexagons within the city of Philadelphia. The analysis showed (available upon request) that for all crime types—other than property crimes, which were rising in the initially treated hexagons in the months prior to the intervention—pre-intervention trends are approximately parallel. To further probe the robustness of our results, we estimated a dynamic lagged panel model that considers the 3 months of crime in each hexagon prior to the immediate effects of upgraded streetlights. The following model was estimated:

$$Y_{it} = \alpha + \beta \text{Lights}_{it} + \sum_{k=1}^{3} \beta_k Y_{it-k} + \delta_i + \pi_t + \epsilon_{it}. \tag{4}$$

This model tests whether there is a significant difference in crime rates after controlling for the prior 3 months of crime, helping us assess whether the parallel trends assumption holds. In the dynamic panel model (8), Y_{it} represents the outcome variable (e.g., crime rate) for hexagon i at month t. The model includes fixed effects for each hexagon θ_i to account for any unobserved, time-invariant characteristics of the individual street segments, and time fixed effects π_t control for any common shocks or trends that affect all hexagons at a given month. The error term, ϵ_{it}, captures any remaining unobserved factors that may influence the outcome variable. By including these lagged values of crime rates, the model tests whether pre-light upgrade trends in crime rates could be driving our estimates of the immediate effects of upgraded streetlights on crime. The results presented in Table 6 are slightly smaller but substantively similar to our primary estimates in Table 2, suggesting that the prior trends in property crime are not substantively impacting the estimates. A difference of coefficients tests across models 1 (Table 2) and 8 (Table 6) shows that they are less than 1.96 standard deviations from each other, confirming the coefficients are not

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TABLE 6 Estimates controlling for 3-month lagged effects.

	Total	Gun	Violent	Property	Nuisance	Financial
Lights	−0.0189***	−0.0046***	−0.0039***	−0.0125***	−0.000402	0.000298
	(0.00301)	(0.000588)	(0.00119)	(0.00244)	(0.000640)	(0.000430)
Ave. Dep	13.53	0.735	3.361	7.497	0.276	0.652
N	14,400	14,400	14,400	14,400	14,400	14,400

Note: Standard errors in parentheses; Hexagon and year-month fixed effects. Three-month lagged estimates included but no shown.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

statistically distinguishable from one another at conventional levels of significance (Clogg et al., 1995).

In order to subject the property crime result—as well as our other findings—to additional scrutiny, we specify two additional models. The first conditions on interacted region-year month fixed effects (Table A3). These models allow each of Philadelphia's four regions to have their own crime trends and therefore account for general changes in crime in different areas of the city. In this model, a treatment effect would only be attributed to the lighting intervention if hexagons with greater exposure to the intervention within the same region experience a reduction in crime. Using an approach recommended by Angrist and Pischke (2009), the second model (Table A4) conditions on hexagon-specific linear time trends, allowing each hexagon to have its own slope, and thus identifies a treatment effect relative to each hexagon's secular trends. Once crime trends are controlled for granularly—through either of the three methods—each of the estimates is very similar to those reported in Table 2.

Next, to determine if our primary results are sensitive to the relatively short 12-month window of pre-intervention data, we repeated the analysis using 2 years of data from before the smart lighting upgrades. The estimates are very similar to those in our original submission and are shown in Table A5.

Finally, we consider whether the estimates are sensitive to measurement error in crime time stamps, which may fail to accurately delineate between daytime and nighttime crimes. The most common sources of error in crime time stamps are that a crime might be written down as having occurred at the time officers arrived on scene or the result of delayed reporting. Crimes such as burglary and theft may experience significant delays in police response or citizen calls for service. To test the robustness of our results, we re-estimated our primary regressions, excluding crimes that occurred within 2 hour of civil twilight. These crimes are most likely to have taken place during daylight hours but are recorded as occurring at night. The revised estimates, presented in Table 7, do not differ substantively from those in Table 2.

3.4 | Qualitative results

Ethnographic observations, focus groups, and stakeholder interviews provided some context for how the street lighting intervention has affected residents' sense of safety, routines, and perceptions of crime prevention. Several observations are worth noting. First, ethnographic observations found numerous nonfunctioning streetlights on blocks with older lights. Some flickered sporadi-

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TABLE 7 Estimates for 2-h buffer around civil twilight hours removed.

	Total	Gun	Violent	Property	Nuisance	Financial
Day	−0.0193***	−0.00283***	−0.00175	−0.0153***	0.000300	0.000394
	(0.00365)	(0.000457)	(0.00119)	(0.00296)	(0.00135)	(0.000401)
Ave. Post	12.12	0.520	2.892	6.769	0.318	0.682
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−5.993	−19.57	−3.880	−8.557	−20.25	−1.579
% Change	−4.376	−14.89	−1.668	−6.207	2.596	1.591
Upper 95%	−2.756	−10.19	0.546	−3.853	25.36	4.755
Night	−0.00938***	−0.00236***	−0.00434***	−0.00552***	0.00000502	−0.0000823
	(0.00146)	(0.000301)	(0.000509)	(0.000785)	(0.000205)	(0.0000832)
Ave. Post	3.222	0.305	1.037	1.377	0.0678	0.0400
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−10.43	−26.37	−14.09	−14.04	−16.02	−16.77
% Change	−7.990	−21.14	−11.47	−10.99	0.204	−5.652
Upper 95%	−5.550	−15.88	−8.840	−7.938	16.42	5.538

*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

cally or gave off only a faint glow, whereas others remained completely dark. There were no signs of intentional damage—no broken glass or visible tampering. Instead, these streetlights appeared to suffer from a lack of maintenance. During twilight observations, the field researcher spoke with residents about the condition of streetlights in their area. Many reported that certain lights had been broken for weeks, and there was a resigned expectation that lighting in their neighborhoods would remain poor, with several commenting they would be "lucky" if a nearby light worked after dark. In contrast, blocks in the same areas with upgraded streetlights had better visibility and no obvious dark areas, as is apparent in Figure 1. The ethnographic observations thus suggest that the efficacy of the improved streetlights is due not only to the better quality of the new lights but to their enhanced "smart" functioning. Notably, LED lights allow city officials to observe an outage right away without the requirement of a resident report to the city's 311 system.

Second, results from focus groups and in-depth interviews suggest that poor lighting enhanced fear and exacerbate existing concerns about crime. Residents in focus groups and interviews noted that they try to stay on streets with good lighting when traveling at night and generally avoid going outside after dark if they can. For example, Sanaa, from Southwest Philadelphia noted:

> I don't go outside after dark… I don't open my door after [dark]; I tell my kids either you're in here or you ain't getting in here… It got really dangerous at a time and we experienced break-ins… Once I became a victim… I had to put a lockdown on my own home…I tell people if you can't come here before [dark], don't bother.

Residents also noted that they would leave their neighborhoods entirely for evening social activities, preferring areas they perceive as safer. Michelle, a Northeast Philadelphia resident, noted:

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> The environment that those [nightlife] places are in is not inviting… People who live in the area, we're familiar with a lot of the bad things that happen [at night] in those environments… So, at nighttime… the majority of people are in their homes.

Third, participants generally agreed that the enhanced street lighting results in crime control gains. Indeed, our subjects demonstrated a sophisticated understanding of the plausible mechanisms associated with the observed crime reductions reported by the impact evaluation. Residents generally viewed lighting improvements as a sign of community investment that could spur informal control and a potential tool for crime deterrence when paired with surveillance cameras. For example, Selena, a Northeast Philadelphia resident, noted:

> I think it would definitely send a message to the community that whoever is upgrading the street lights actually cares…they care about jobs…about our community… and about our welfare and our well-being. I think it would also foster maybe a beginning or a starting point of people being able to feel just a little bit more comfortable outside during nighttime hours, so eventually we could get to that point, one day, where a lot more people could be outside during the nighttime hours if they wanted to and not just have to stay stayed locked in their homes just because they're scared.

Selena and other participants suggested lighting improvements signaled municipal investments that could serve as a catalyst for renewed community engagement and subsequent collective action. They were hopeful that continued improvements to the neighborhood might rebuild local collective efficacy that eroded through time due to neglect and a lack of investment.

Most participants also believed that improved street lighting modified the physical environment in ways that diminished easy crime opportunities. Subjects noted that increasing visibility reduced the ability of prospective offenders to conceal themselves in "hiding spots" after dark. Marcus, a North Philadelphia resident, explained, "if it's bright and they're visible, they're not just gonna be standing there." Damon highlighted the link between improved lighting, surveillance, and deterrence by observing that [prospective shooters] "don't want to be using their guns in the light" where "they can easily be seen." Terrell speculated the lighting would improve the quality of videos captured by surveillance cameras in the area and, in turn, facilitate the identification of offenders because "more people could see" who committed the crime.

Although residents noted the importance of lighting in navigating streets at night and most reported an enhanced sense of personal safety, some expressed skepticism about the ability of upgraded lighting to curb gun violence. These skeptics emphasized that shootings often occurred in broad daylight, suggesting lighting was insufficient to address the types of systemic issues that give rise to endemic violence. This sentiment was expressed clearly by two separate residents. Quinta, a West Philadelphia resident, opined "upgrading the street lighting is not going to make me feel safe outside my door… not when they shoot all day… They are literally shooting outside my door now." Similarly, Jason, a North Philadelphia resident stated "around here, a lot of shootings transpire in broad daylight… I had a guy who died in my hands, brains coming out the back of his head, this happened in broad daylight like 1:00 in the afternoon…"

Residents appreciated improved streetlights as a step in the right direction but emphasized the need for broader structural changes to make large and lasting impacts on public safety in their neighborhood. Enhanced streetlights were essential to reducing fear and enhancing safety during the nighttime. However, improved street lighting as a standalone intervention was viewed as a superficial fix relative to the prospects of making long-term investments to develop local

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employment opportunities, improve poorly functioning schools, support impoverished families, and provide resources to at-risk youth.

4 | DISCUSSION

This study reports results for the first 10 months of a citywide upgrade of streetlights on crime in Philadelphia. The new LED streetlights provide better visibility by emitting white lighting, as opposed to yellow-orange light on existing HPS bulbs, and lead to a reduction in "dark spots" in affected communities. We find that outdoor nighttime street crimes declined by 15% and outdoor nighttime gun violence declined by 21% as a function of the enhanced lighting. We estimate that the intervention reduced overall crimes by approximately 5%. The crime reductions are considerably smaller during daytime hours but are still present, suggesting that the benefits of the intervention diffused to the daytime hours, a possibility that has been noted in prior reviews of the literature (Welsh & Farrington, 2008; Welsh et al., 2022), as well as the long-term follow-up from the literature's role randomized experiment (Mitre-Becerril et al., 2022). With nearly a third of street segments in Philadelphia experiencing lighting upgrades, we were able to assess the benefits of smart lighting on a major scale. The investment in better lighting infrastructure appears to contribute to meaningful reductions in crime above and beyond the overall citywide trends. Regarding whether enhanced lighting merely displaces crime, consistent with the findings of Chalfin et al. (2022) and Tompson et al. (2023), we observe that when a block is upgraded with enhanced LED lights, crime decreases slightly in adjacent street segments, indicating a small diffusion of benefits.

Next, we examine the crime-related social cost savings resulting from investments in upgraded smart street lighting in Philadelphia. Using estimates from the RAND costs of crime calculator, we calculate a unit social cost of crime of \$7490 for April 2024.[5] During an average follow-up period of 3.28 months, we estimate a reduction of 0.0287 fewer crimes per month per kilometer, which totals approximately 0.09 fewer crimes (-0.0287×3.28) in total. This translates to a monthly reduction of \$697 in crime costs due to the investment in smart lighting.

The total cost of the lighting upgrades was approximately \$91 million, covering 134,550 lights across the entire City of Philadelphia. This amounts to roughly \$676 per light (\$91,000,000 ÷ 134,550 = \$675.93). During the observation period, we recorded an average of 11.14 lights per kilometer per month in a hexagon, or an average of \$7531 per month (11.14 × \$676 = \$7531.04). As a result, the investment in upgraded street lighting may have yielded a social benefit of 9% in crime reduction (\$697 ÷ \$7531.04 = 9.2%). The social benefit from crime reduction would be even greater if we focus specifically on gun offenses, which have an average social cost of \$126,234 (using aggravated assaults as a proxy). With an estimated reduction of 0.00519 in gun offenses per month, this results in a social benefit of \$2150 (-0.00519×3.28 months × \$126,234). Based on this calculation, the lighting upgrades could have generated a 29% social benefit from the reduction in gun violence alone (\$2150 ÷ \$7531.04 = 28.5%). The smart lighting upgrades were financed through the City of Philadelphia's "sustainability bonds," which are expected to offset their installation costs through energy savings through time. We note that, even within a very short time window, the additional crime reduction benefits represent an unexpected bonus beyond the originally anticipated energy savings.

Although the magnitude of our results aligns broadly with the central estimates in prior literature, their significance is different because they reflect an impact on a major scale in a U.S. city suffering from serious crime problems. Previous studies have shown that improved lighting can

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reduce crime in specific areas, such as neighborhoods or public housing developments. However, our research demonstrates that Philadelphia's investment in upgrading 34,374 streetlights has had a measurable effect on the city's overall crime rate. To clarify, this study focused on upgrades in areas that account for 30.7% of the city's gun crimes. Given the 31% decline in gun violence observed during the study period, Philadelphia's lighting investment may account for as much as one sixth of the citywide decline.

Our qualitative analysis suggests that these crime control gains were achieved without causing any unintended harm to area residents. Respondents generally perceived that their neighborhoods were safer and reported increased use of public spaces after the enhanced LED lights were installed. Given concerns about proactive policing strategies that expose larger shares of residents to the criminal justice system, generate racial disparities, and risk stimulating abusive police–citizen encounters (e.g., see Alexander, 2010; Hinton, 2016; Kohler-Hausmann, 2018), enhanced street lighting is an effective situational crime prevention measure that avoids these unintended consequences of enforcement actions. LED streetlight installations fit with community demand for alternative violence prevention measures following the post-COVID-19 increases in gun violence and contemporaneous social justice protests after the tragic murder of George Floyd at the hands of Minneapolis police officers. Improved street lighting programs can be viewed as part of public health and urban planning community violence interventions that seek to prevent crime by changing places through blight remediation, greening vacant lots, and razing abandoned buildings (MacDonald et al., 2019; Webster, 2022).

Evidence suggests that some social service-based strategies can effectively reduce violence among at-risk youth (Botvin et al., 2006; Heller et al., 2017). Unfortunately, scaling social service interventions to reduce serious crime at the community level has proven challenging. One major difficulty is that effective interventions are often hard to expand (Davis et al., 2017). Several factors contribute to this, particularly the diminishing returns on staff productivity as organizations grow and the challenge of expanding an organization originally built by a visionary leader, especially when leadership of this quality may be scarce. Even well-established and effective citywide violence prevention programs, such as focused deterrence programs to reduce gang violence (Braga & Kennedy, 2021; Braga et al., 2008), have collapsed when productive managers and agency executives moved on from their current positions. Scaling up and sustaining effective violence prevention programs are often challenged by the limits of human capital in some cities.

A key advantage of investments like enhanced street lighting is that they do not require charismatic leadership, heroic mid-level managers, or rare expertise for installation or maintenance. Additionally, our interviews with community members suggest that residents view upgraded lighting as a positive sign of city investment, even if they did not think lighting on its own was a sufficient crime prevention approach. Although the fixed costs of lighting installations can be high, the variable costs of maintaining them are relatively low—likely no higher than the costs associated with maintaining legacy lighting, and potentially even lower. As a result, street lighting appears to be a scalable policy that can be used in conjunction with other programs to help reduce crime, as demonstrated by cities like Chicago, Dallas, Los Angeles, and Philadelphia, which have already made such investments.

4.1 | Limitations

A key strength of this study is its ability to test whether upgraded street lighting reduces crime on a citywide scale. However, several limitations warrant discussion. First, the study was conducted

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in Philadelphia, a large, dense, and pedestrian-oriented city. The effects of upgraded lighting may differ in cities with lower density or those that are more car dependent. Second, although the study has sufficient statistical power to detect short-run effects during an average exposure period of 3.28 months, it is not powered to estimate the impact of the timing of installations or to assess long-term effects. Notably, Mitre-Becerril et al. (2022) find evidence that lighting effects may persist for up to 3 years after installation. Third, the installation rollout was not randomized. Early installation areas had higher baseline crime, raising the possibility of correlation with other citywide improvements. Although this cannot be entirely ruled out, the estimates remain robust when we estimated a model with region-by-month fixed effects and hexagon-specific linear trends. Moreover, modeling lighting as a continuous measure of upgraded lights reduces the likelihood that confounding factors followed the same pattern. There is also the possibility that enhanced lighting may have attracted more police to certain blocks. However, this is unlikely, given that police deployment is determined by calls for service and strategic deployment to operational areas. Finally, although this study is among the best-powered of existing studies, it remains underpowered to detect heterogeneous effects across different urban features, especially after correcting *p*-values for multiple hypothesis testing. Like prior studies, we cannot determine the mechanisms through which upgraded street lighting reduces crime. One promising avenue for future research is to explore whether upgraded lighting alters how residents use public space, potentially by leveraging mobility data from cell phones.

FUNDING INFORMATION

Research reported in this publication was supported by the National Institute on Minority Health and Health Disparities of the National Institutes of Health under award number R01MD019163.

ACKNOWLEDGMENTS

The authors would like to thank Devin Haskins, Monica Rodriguez, Duane Thornton, and Arun Prabhakaran at the Urban Affairs Coalition for their collaboration and organization of focus groups and key informant interviews. We also extend our sincere thanks to Katie Bartolotta and the staff at the Philadelphia Energy Authority for generously sharing their data and to Michael Fuller of Keystone Lighting Solutions for his insightful contributions regarding the data and the technical aspects of the light upgrades.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ENDNOTES

[1] The "smart" aspect of this intervention is that all newly- and already-converted LED streetlights are linked to a lighting management system, through which lighting brightness can be remotely controlled and outages can be monitored (phillystreetlightimprovement.com/).

[2] policysearch.ama-assn.org/policyfinder/detail/H-135.927?uri=%2FAMADoc%2FHOD-135.927.xml

[3] We also calculated the raw counts of crime and lighting per month in each hexagon. The Spearman rank correlations were greater than 0.95 between counts and kernel density estimates. Estimates were similar using counts rather than rates, but sensitive to extreme count outliers in the top 99th percentile of the crime distribution. Kernel density estimates have fewer outliers because counts are weighted by distance from the centroid of each hexagon. We prefer kernel density estimates because it provides a more accurate account of the exact spatial density of light and crime locations. Results of models with counts per hexagon are displayed in Table A1.

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[4]The estimates work according to the following equation based on quartic function: $\hat{f}(X,Y) = \frac{1}{n\times 1000^2}\sum_{i=1}^{n}\frac{15}{16}\left(1-\left(\frac{\sqrt{(X-x_i)^2+(Y-y_i)^2}}{1000}\right)^2\right)^2$ for $\left|\frac{\sqrt{(X-x_i)^2+(Y-y_i)^2}}{1000}\right| \leq 1$. For each hexagon center point (X, Y), the weighted number of crime or light points (i) (x, y) is calculated based on the Euclidean distance.

[5]rand.org/well-being/justice-policy/centers/quality-policing/cost-of-crime.html

REFERENCES

Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American statistical Association*, *105*(490), 493–505.

Alexander, M. (2010). *The new Jim Crow*. The New Press.

Angrist, J. D., & Pischke, J. S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton University Press.

Arvate, P., Falsete, F. O., Ribeiro, F. G., & Souza, A. P. (2018). Lighting and homicides: Evaluating the effect of an electrification policy in rural Brazil on violent crime reduction. *Journal of Quantitative Criminology*, *34*(4), 1047–1078.

Benartzi, S., & Thaler, R. H. (2013). Behavioral economics and the retirement savings crisis. *Science*, *339*(6124), 1152–1153.

Botvin, G., Griffin, K., & Nichols, T. (2006). Preventing youth violence and delinquency through a universal school-based prevention approach. *Prevention Science*, *7*(4), 403–408.

Bowers, K., & Guerette, R. (2014). Effectiveness of situational crime prevention. In G. Bruinsma & D. Weisburd (Eds.), *Encyclopedia of criminology and criminal justice* (pp. 1318–1329). Springer.

Braga, A. (2008). *Problem-oriented policing and crime prevention* (2nd ed.). Criminal Justice Press.

Braga, A., & Bond, B. (2008). Policing crime and disorder hot spots: A randomized controlled trial. *Criminology*, *46*(3), 577–607.

Braga, A., & Clarke, R. (2014). Explaining high risk concentrations of crime in the city. *Journal of Research in Crime and Delinquency*, *51*(4), 480–498.

Braga, A., Hureau, D., & Winship, C. (2008). Losing faith? Police, black churches, and the resurgence of youth violence in Boston. *Ohio State Journal of Criminal Law*, *6*(1), 141–172.

Braga, A., & Kennedy, D. (2021). *A framework for addressing violence and serious crime*. Cambridge University Press.

Braga, A., Papachristos, A., & Hureau, D. (2010). The concentration and stability of gun violence at micro places in Boston, 1980–2008. *Journal of Quantitative Criminology*, *26*, 33–53.

Braithwaite, J. (2021). Scaling up crime prevention and justice. *Crime and Justice*, *50*, 247–299.

Branas, C., Culhane, D., Richmond, T., & Wiebe, D. (2008). Novel linkage of individual and geographic data to study firearm violence. *Homicide Studies*, *12*(3), 298–320.

Branas, C., Kondo, M., Murphy, S., South, E., Polsky, D., & MacDonald, J. (2016). Urban blight remediation as a cost-beneficial solution to firearm violence. *American Journal of Public Health*, *106*(12), 2158–2164.

Branas, C. C., Cheney, R. A., Macdonald, J. M., Tam, V. W., Jackson, T. D., & Ten Have, T. R. (2011). A difference-in-differences analysis of health, safety, and greening vacant urban space. *American Journal of Epidemiology*, *174*(11), 1296–1306.

Branas, C. C., & MacDonald, J. M. (2014). A simple strategy to transform health, all over the place. *Journal of Public Health Management and Practice*, *20*(2), 157–159.

Branas, C. C., South, E., Kondo, M. C., Hohl, B. C., Bourgois, P., Wiebe, D. J., & Macdonald, J. M. (2018). Citywide cluster randomized trial to restore blighted vacant land and its effects on violence, crime, and fear. *Proceedings of the National Academy of Sciences of the United States of America*, *115*(12), 2946–2951. https://doi.org/10.1073/pnas.1718503115

Brantingham, P. J., & Brantingham, P. L. (Eds.). (1991). *Environmental criminology* (2nd ed.). Waveland Press.

Bursik, R., & Grasmick, H. (1993). *Neighborhoods and crime*. Lexington Books.

Casteel, C., & Peek-Asa, C. (2000). Effectiveness of crime prevention through environmental design (CPTED) in reducing robberies. *American Journal of Preventive Medicine*, *18*(4), 99–115.

17459133, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1745-9133.70006 by University Of Chicago Library, Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chalfin, A., Hansen, B., Lerner, J., & Parker, L. (2021). Reducing crime through environmental design: Evidence from a randomized experiment of street lighting in New York City. *Journal of Quantitative Criminology*, *38*, 127–157.

Chalfin, A., Kaplan, J., & LaForest, M. (2022). Street light outages, public safety, and crime attraction. *Journal of Quantitative Criminology*, *38*(4), 891–919.

Clarke, R. (Ed.). (1997). *Situational crime prevention*. Harrow and Heston.

Clogg, C. C., Petkova, E., & Haritou, A. (1995). Statistical methods for comparing regression coefficients between models. *American Journal of Sociology*, *100*(5), 1261–1293.

Cohen, L. E., & Felson, M. (1979). Social change and crime rate trends. *American Sociological Review*, *44*, 588–605.

Cornish, D. B., & Clarke, R. V. (1986). *The reasoning criminal*. Springer-Verlag.

Cozens, P., Saville, G., & Hillier, D. (2005). Crime prevention through environmental design (CPTED): A review and modern bibliography. *Property Management*, *25*(5), 328–356.

Davis, J. M., Guryan, J., Hallberg, K., & Ludwig, J. (2017). *The economics of scale-up* (No. w23925). National Bureau of Economic Research.

de Chaisemartin, C., & D'Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, *110*(9), 2964–2996.

de Chaisemartin, C., D'Haultfœuille, X., & Guyonvarch, Y. (2019). Fuzzy differences-in-differences with Stata. *The Stata Journal*, *19*(2), 435–458.

Del Pozo, B., Knorre, A., Mello, M. J., & Chalfin, A. (2022). Comparing risks of firearm-related death and injury among young adult males in selected US cities with wartime service in Iraq and Afghanistan. *JAMA Network Open*, *5*(12), Article e2248132.

Diez Roux, A. V., & Mair, C. (2010). Neighborhoods and health. *Annals of the New York Academy of Sciences*, *1186*(1), 125–145.

Doleac, J. L., & Sanders, N. (2015). Under the cover of darkness: How ambient light influences criminal activity. *Review of Economics and Statistics*, *97*(5), 1093–1103.

Dominguez, P., & Asahi, K. (2017). Crime time: How ambient light affects criminal activity. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.5150459

Dossey, L. (2019). Night in the big city: Temporal patterns in Antioch and Constantinople as revealed by Chrysostom's sermons. In C. de Wet & W. Mayer (Eds.), *Revisioning John Chrysostom* (pp. 698–732). Brill.

Farrington, D., & Welsh, B. (2002). *Effects of improved street lighting on crime: A systematic review*. London Home Office.

Garvin, E. C., Cannuscio, C. C., & Branas, C. (2012). Greening vacant lots to reduce violent crime: A randomized controlled trial. *Injury Prevention*, *19*(3), 198–203.

Goldstein, H. (1990). *Problem-oriented policing*. Temple University Press.

Guerette, R. T., & Bowers, K. J. (2017). Assessing the extent of crime displacement and diffusion of benefits: A review of situational crime prevention evaluations. In M. Natarajan (Ed.), *Crime opportunity theories* (pp. 529–566). Routledge.

Heller, S. B., Shah, A. K., Guryan, J., Ludwig, J., Mullainathan, S., & Pollack, H. A. (2017). Thinking, fast and slow? Some field experiments to reduce crime and dropout in Chicago. *The Quarterly Journal of Economics*, *132*(1), 1–54.

Hinton, E. (2016). *From the war on poverty to the war on crime*. Harvard University Press.

Jacobs, B. A. (2010). Deterrence and deterrability. *Criminology*, *48*(2), 417–441.

St Jean, P. (2007). *Pockets of crime*. University of Chicago Press.

Jeffery, C. (1977). *Crime prevention through environmental design*. Sage Publications.

Kaplan, J., & Chalfin, A. (2021). Ambient lighting, use of outdoor spaces and perceptions of public safety: Evidence from a survey experiment. *Security Journal*, *35*(3), 694–724.

Kim, I. T., Jang, I. H., Choi, A. S., & Sung, M. (2015). Brightness perception of white LED lights with different correlated colour temperatures. *Indoor and Built Environment*, *24*(4), 500–513.

Kohler-Hausmann, I. (2018). *Misdemeanorland*. Princeton University Press.

Kondo, M. C., Fluehr, J. M., McKeon, T., & Branas, C. C. (2018). Urban green space and its impact on human health. *International Journal of Environmental Research and Public Health*, *15*(3), Article 445.

Kondo, M. C., Keene, D., Hohl, B. C., MacDonald, J. M., & Branas, C. C. (2015). A difference-in-differences study of the effects of a new abandoned building remediation strategy on safety. *PLoS ONE*, *10*(7), Article e0129582.

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Lee, D. S., & McCrary, J. (2017). The deterrence effect of prison: Dynamic theory and evidence. *Regression Discontinuity Designs*, *38*, 73–146.

Ludwig, J., Kling, J. R., & Mullainathan, S. (2011). Mechanism experiments and policy evaluations. *Journal of Economic Perspectives*, *25*(3), 17–38.

MacDonald, J., Branas, C., & Stokes, R. (2019). *Changing places*. Princeton University Press.

MacDonald, J., Mohler, G., & Brantingham, P. (2022). Association between race, shooting hot spots, and the surge in gun violence during the COVID-19 pandemic in Philadelphia, New York, and Los Angeles. *Preventive Medicine*, *165*, Article 107241.

MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, *35*(1), 13–39.

Madrian, B. C., & Shea, D. F. (2001). The power of suggestion: Inertia in 401(k) participation and savings behavior. *The Quarterly Journal of Economics*, *116*(4), 1149–1187.

Mitre-Becerril, D., Tahamont, S., Lerner, J., & Chalfin, A. (2022). Can deterrence persist? Long-term evidence from a randomized experiment in street lighting. *Criminology & Public Policy*, *21*(4), 865–891.

Nagin, D. S. (2013). Deterrence in the twenty-first century. *Crime and Justice in America*, *42*(1), 199–263.

Nagin, D. S., & Paternoster, R. (1994). Personal capital and social control: The deterrence implications of a theory of individual differences in criminal offending. *Criminology*, *32*(4), 581–606.

Nagin, D. S., Solow, R. M., & Lum, C. (2015). Deterrence, criminal opportunities, and police. *Criminology*, *53*(1), 74–100.

Newman, O. (1972). *Defensible space: Crime prevention through urban design*. Macmillan.

Painter, K. (1996). The influence of street lighting improvements on crime, fear and pedestrian street use, after dark. *Landscape and Urban Planning*, *35*(2-3), 193–201.

Painter, K., & Farrington, D. P. (1997). The crime reducing effect of improved street lighting: The Dudley project. In R. V. Clarke (Ed.), *Situational crime prevention*: S*uccessful case studies* (2nd ed., pp. 209–226). Harrow and Heston.

Ratcliffe, J. (2010). Crime mapping: Spatial and temporal challenges. In A. R. Piquero & D. Weisburd (Eds.), *Handbook of quantitative criminology* (pp. 5–24). Springer.

Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, *4*(3), 305–322.

Saldaña, J. (2015). *Thinking qualitatively*. Sage Publications.

Sampson, R. J., Raudenbush, S. W., & Earls, F. (1997). Neighborhoods and violent crime: A multilevel study of collective efficacy. *Science*, *227*(5328), 918–924.

Sampson, R. J., Wilson, W. J., & Katz, H. (2018). Reassessing "Toward a theory of race, crime, and urban inequality": Enduring and new challenges in 21st century America. *Du Bois Review: Social Science Research on Race*, *15*(1), 13–34.

Sherman, L. W., Gartin, P. R., & Buerger, M. E. (1989). Hot spots of predatory crime. *Criminology*, *27*(1), 27–56.

South, E., MacDonald, J., Tam, V., Ridgeway, G., & Branas, C. (2023). Effect of abandoned housing interventions on gun violence, perceptions of safety, and substance use in Black neighborhoods: A citywide cluster randomized trial. *JAMA Internal Medicine*, *183*(1), 31–39.

Taylor, R. B., & Gottfredson, S. (1986). Environmental design, crime, and prevention: An examination of community dynamics. *Communities and Crime*, *8*, 387–416.

Thaler, R. (2018). From cashews to nudges: The evolution of behavioral economics. *American Economic Review*, *108*(6), 1265–1287.

Thaler, R., & Sunstein, C. (2009). *Nudge: Improving decisions about health, wealth, and happiness*. Penguin Books.

Tompson, L., Steinbach, R., Johnson, S. D., Teh, C. S., Perkins, C., Edwards, P., & Armstrong, B. (2023). Absence of street lighting may prevent vehicle crime, but spatial and temporal displacement remains a concern. *Journal of Quantitative Criminology*, *39*(3), 603–623.

Webster, D. W. (2022). Public health approaches to reducing community gun violence. *Daedalus*, *151*(1), 38–48.

Weisburd, D., Bushway, S., Lum, C., & Yang, S. (2004). Trajectories of crime at places. *Criminology*, *42*(2), 283–322.

Weisburd, D., Groff, E., & Yang, S. (2012). *The criminology of place*. Oxford University Press.

Weisburd, D., Wyckoff, L. A., Ready, J., Eck, J. E., Hinkle, J. C., & Gajewski, F. (2006). Does crime just move around the corner? A controlled study of spatial displacement and diffusion of crime control benefits. *Criminology*, *44*(3), 549–592.

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Welsh, B., & Farrington, D. (2008). Effects of improved street lighting on crime. *Campbell Systematic Reviews*, *4*(1), 1–51.

Welsh, B., & Farrington, D. (2009). *Making public places safer*. Oxford University Press.

Welsh, B., Farrington, D., & Douglas, S. (2022). The impact and policy relevance of street lighting for crime prevention: A systematic review based on a half-century of evaluation research. *Criminology & Public Policy*, *21*(3), 739–765.

Welsh, B., Sullivan, C., & Olds, D. (2010). When early crime prevention goes to scale. *Prevention Science*, *11*(2), 115–125.

Wheeler, A. (2023). Analysis of LED street light conversions on firearm crimes in Dallas, Texas. *CrimRxiv*. https://doi.org/10.21428/cb6ab371.c8186ce7

Wilcox, P., & Cullen, F. T. (2018). Situational opportunity theories of crime. *Annual Review of Criminology*, *1*(1), 123–148.

Wright, R., & Decker, S. (1997). *Armed robbers in action*. Northeastern University Press.

How to cite this article: MacDonald, J. M., Chalfin, A., Moritz, M., Wade, B., Mendlein, A. K., Braga, A. A., & South, E. (2026). Can enhanced street lighting improve public safety at scale?. *Criminology & Public Policy*, *25*, 31–62. https://doi.org/10.1111/1745-9133.70006

APPENDIX

TABLE A1 Estimates from counts of crime per hexagon.

	Total	Gun	Violent	Property	Nuisance	Financial
All	−0.024***	−0.00488***	−0.00512**	−0.0157***	0.000648	0.000354
	(0.00459)	(0.000855)	(0.00171)	(0.00377)	(0.00188)	(0.000541)
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−7.161	−25.84	−7.026	−9.291	−25.36	−3.215
% Change	−5.229	−19.26	−4.251	−6.325	5.433	1.610
Upper 95%	−3.295	−12.67	−1.474	−3.356	36.09	6.429
Inside	0.00667	−0.00122***	0.000342	0.00511	0.000470	0.000414
	(0.00347)	(0.000333)	(0.00105)	(0.00321)	(0.000325)	(0.000479)
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−0.0656	−32.76	−2.497	−1.000	−9.619	−2.696
% Change	3.042	−21.41	0.500	4.315	27.12	2.129
Upper 95%	6.146	−10.01	3.495	9.618	63.28	6.947
Outside	−0.034***	−0.00384***	−0.00613***	−0.0210***	−0.000223	0.0000145
	(0.00315)	(0.000742)	(0.00111)	(0.00219)	(0.00147)	(0.000153)
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−18.86	−28.47	−16.48	−19.53	−32.68	−13.70
% Change	−16.03	−20.69	−12.17	−16.24	−2.353	0.699
Upper 95%	−13.20	−12.88	−7.857	−12.94	28.03	15.09
Daytime	−0.0112**	−0.00205***	0.000681	−0.00993**	0.000494	0.000240
	(0.00372)	(0.000521)	(0.00126)	(0.00327)	(0.00132)	(0.000509)

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TABLE A1 (Continued)

	Total	Gun	Violent	Property	Nuisance	Financial
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−5.778	−24.78	−2.360	−8.905	−26.74	−3.966
% Change	−3.506	−16.57	0.900	−5.418	6.318	1.258
Upper 95%	−1.231	−8.322	4.157	−1.925	39.20	6.477
Outside: Daytime	−0.022***	−0.00147**	−0.00137	−0.0155***	−0.000235	0.0000137
	(0.00250)	(0.000456)	(0.000754)	(0.00187)	(0.000981)	(0.000143)
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−18.60	−27.48	−9.316	−19.99	−35.48	−14.90
% Change	−15.32	−17.14	−4.495	−16.18	−3.900	0.769
Upper 95%	−12.02	−6.750	0.337	−12.37	27.78	16.43
Nighttime	−0.013***	−0.00279***	−0.00576***	−0.00564***	0.000142	0.000121
	(0.00210)	(0.000510)	(0.000892)	(0.00127)	(0.000600)	(0.000159)
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−11.90	−28.69	−16.41	−12.18	−24.64	−6.372
% Change	−9.056	−21.16	−12.60	−8.461	3.400	4.076
Upper 95%	−6.207	−13.61	−8.782	−4.739	31.35	14.49
Nighttime: Outside	−0.012***	−0.00232***	−0.00469***	−0.00525***	−0.00000377	0.00000585
	(0.00114)	(0.000441)	(0.000672)	(0.000666)	(0.000535)	(0.0000484)
N	16,560	16,560	16,560	16,560	16,560	16,560
Lower 5%	−20.81	−31.11	−29.59	−18.85	−29.79	−28.62
% Change	−17.58	−22.71	−23.15	−15.11	−0.107	1.895
Upper 95%	−14.34	−14.28	−16.68	−11.36	29.58	32.36

Note: Standard errors in parentheses; Hexagon and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

TABLE A2 Estimates with street segments per light added.

No. Lights	Total	Gun	Violent	Property	Nuisance	Financial
1	1.034	0.901	1.024	1.051	1.045	0.989
	(0.0541)	(0.0809)	(0.0501)	(0.0874)	(0.217)	(0.0976)
2	1.126	0.965	1.052	1.163	1.178	1.175
	(0.0723)	(0.0973)	(0.0554)	(0.117)	(0.196)	(0.136)
3	1.047	0.979	1.164**	0.966	0.848	1.307*
	(0.0703)	(0.110)	(0.0646)	(0.112)	(0.166)	(0.149)
4	0.985	1.054	1.091	0.907	0.842	1.457**
	(0.0478)	(0.122)	(0.0709)	(0.0620)	(0.209)	(0.186)
5	1.098	1.165	1.151*	1.052	0.773	1.320*
	(0.0544)	(0.148)	(0.0801)	(0.0783)	(0.192)	(0.180)

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TABLE A2 (Continued)

No. Lights	Total	Gun	Violent	Property	Nuisance	Financial
6	0.970	0.904	1.040	0.869	0.771	1.122
	(0.0921)	(0.139)	(0.135)	(0.0856)	(0.216)	(0.171)
7	0.796**	0.968	0.736**	0.813	0.724	1.170
	(0.0694)	(0.173)	(0.0859)	(0.0946)	(0.306)	(0.212)
8	0.827	1.069	0.915	0.798	0.865	0.708
	(0.118)	(0.185)	(0.0895)	(0.151)	(0.358)	(0.158)
9	0.748*	0.971	0.787	0.666**	0.406	0.915
	(0.0911)	(0.220)	(0.110)	(0.0871)	(0.239)	(0.223)
10+	0.862	0.882	0.710***	0.958	0.882	0.797
	(0.0871)	(0.113)	(0.0599)	(0.140)	(0.190)	(0.129)
Ave. Dep	0.368	0.0710	0.144	0.239	0.106	0.0668
N	499,100	125,787	307,786	441,416	29,992	143,865

Note: Standard errors in parentheses; Hexagon and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

TABLE A3 Estimates with region time fixed effects.

	Total	Gun	Violent	Property	Nuisance	Financial
All	−0.0415***	−0.00614***	−0.0101***	−0.0304***	0.00113	−0.000350
	(0.00425)	(0.000654)	(0.00141)	(0.00307)	(0.00167)	(0.000408)
N	15,796	15,796	15,796	15,796	15,796	15,796
Lower 5%	−8.189	−22.12	−8.280	−11.25	−13.51	−3.961
% Change	−6.825	−18.32	−6.497	−9.395	7.153	−1.207
Upper 95%	−5.459	−14.51	−4.713	−7.536	27.63	1.552
Day	−0.0366***	−0.00342***	−0.00533***	−0.0268***	0.000240	−0.000866*
	(0.00354)	(0.000378)	(0.00109)	(0.00265)	(0.00105)	(0.000387)
N	15,796	15,796	15,796	15,796	15,796	15,796
Lower 5%	−10.63	−26.41	−7.877	−13.74	−17.90	−6.538
% Change	−8.943	−21.75	−5.634	−11.52	2.377	−3.490
Upper 95%	−7.256	−17.08	−3.386	−9.302	22.58	−0.435
Night	−0.00490**	−0.00270***	−0.00473***	−0.00353**	0.000885	0.000519***
	(0.00186)	(0.000387)	(0.000654)	(0.00116)	(0.000131)	(0.000131)
N	15,796	15,796	15,796	15,796	15,796	15,796
Lower 5%	−4.291	−19.46	−9.952	−6.422	−6.669	6.245
% Change	−2.460	−15.22	−7.836	−3.913	15.53	12.30
Upper 95%	−0.625	−10.97	−5.717	−1.399	37.38	18.34

Note: Standard errors in parentheses; Hexagon and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

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TABLE A4 Estimates with hexagon linear time trends.

	Total	Gun	Violent	Property	Nuisance	Financial
All	−0.0584***	−0.00880***	−0.0147***	−0.0420***	0.00357	−0.000657
	(0.00669)	(0.00123)	(0.00243)	(0.00507)	(0.00323)	(0.000684)
N	9736	9736	9736	9736	9736	9736
Lower 5%	−11.72	−33.27	−12.54	−16.04	−17.38	−6.878
% Change	−9.581	−26.20	−9.490	−12.99	22.48	−2.266
Upper 95%	−7.436	−19.09	−6.432	−9.932	61.34	2.360
Day	−0.0714***	−0.00649***	−0.0140***	−0.0482***	0.000861	−0.00145*
	(0.00662)	(0.000778)	(0.00190)	(0.00475)	(0.00190)	(0.000638)
N	9736	9736	9736	9736	9736	9736
Lower 5%	−20.51	−50.40	−18.65	−24.59	−28.12	−10.85
% Change	−17.38	−40.96	−14.75	−20.64	8.509	−5.831
Upper 95%	−14.24	−31.46	−10.84	−16.68	44.71	−0.797
Night	0.0131***	−0.00229**	−0.000696	0.00618**	0.00271	0.000792**
	(0.00331)	(0.000715)	(0.00134)	(0.00183)	(0.00146)	(0.000219)
N	9736	9736	9736	9736	9736	9736
Lower 5%	3.308	−20.75	−5.500	2.861	−2.632	8.635
% Change	6.555	−12.91	−1.157	6.831	46.93	18.75
Upper 95%	9.795	−5.019	3.192	10.79	94.59	28.78

Note: Standard errors in parentheses; Hexagon and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

TABLE A5 Estimates with 2 years of preperiod.

	Total	Gun	Violent	Property	Nuisance	Financial
All	−0.0207***	−0.00661***	−0.00967***	−0.0155***	0.000919	0.000654
	(0.00435)	(0.000745)	(0.00151)	(0.00310)	(0.00166)	(0.000511)
N	24,480	24,480	24,480	24,480	24,480	24,480
Lower 5%	−4.793	−24.41	−8.123	−6.676	−14.66	−1.200
% Change	−3.395	−20.03	−6.222	−4.800	5.799	2.261
Upper 95%	−1.995	−15.63	−4.318	−2.921	26.10	5.714
Day	−0.0166***	−0.00352***	−0.00431***	−0.0122***	−0.0000302	0.000241
	(0.00366)	(0.000415)	(0.00115)	(0.00271)	(0.00108)	(0.000479)
N	24,480	24,480	24,480	24,480	24,480	24,480
Lower 5%	−5.796	−27.70	−6.923	−7.508	−21.10	−2.819
% Change	−4.047	−22.54	−4.551	−5.230	−0.298	0.973
Upper 95%	−2.296	−17.37	−2.174	−2.948	20.51	4.759
Night	−0.00407	−0.00307***	−0.00534***	−0.00333*	0.000945	0.000419**
	(0.00211)	(0.000433)	(0.000717)	(0.00116)	(0.000642)	(0.000150)

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TABLE A5 (Continued)

	Total	Gun	Violent	Property	Nuisance	Financial
N	24,480	24,480	24,480	24,480	24,480	24,480
Lower 5%	−4.113	−22.50	−11.11	−6.207	−5.518	2.966
% Change	−2.044	−17.67	−8.803	−3.696	16.52	9.956
Upper 95%	0.0288	−12.82	−6.493	−1.180	38.20	16.91

Note: Standard errors in parentheses; Hexagon and year-month fixed effects.
*$p < 0.05$;
**$p < 0.01$;
***$p < 0.001$.

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